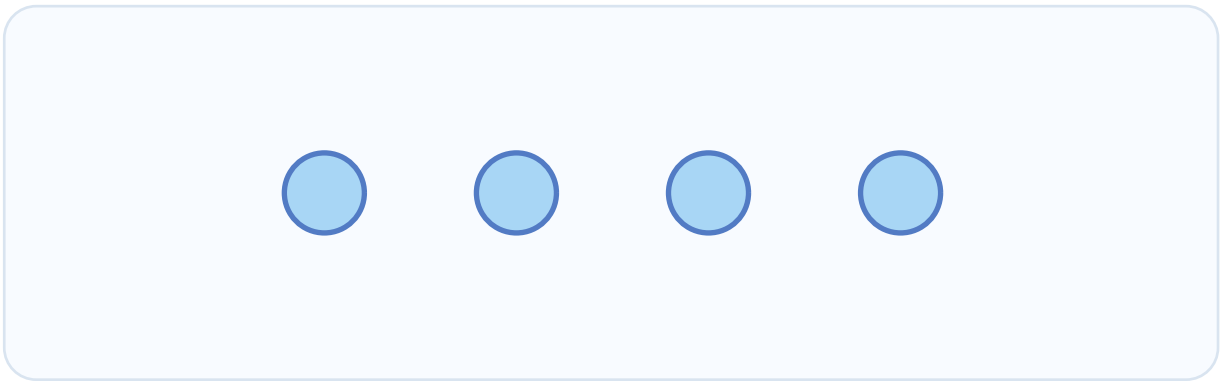
1. How to use the library



The new functions return drawing instructions. Give each picture a name, then put it in image_sequence. Run student_sequence.py with F5.

Try this in student_images.py

```
from turtle_shapes import *

my_triangle = triangle(x=-2, y=-2, size=3,
                       color="blue")
my_star = star(x=1, y=1, size=2,
               color="orange")

image_sequence = [
    ("Triangle", my_triangle),
    ("Star", my_star),
]
```

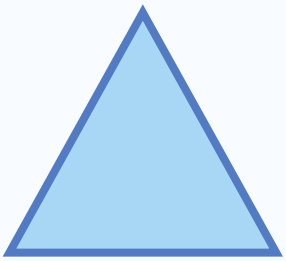
Remember

Use the same variable name in both places. Coordinates usually fit between -5 and 5. The functions do not draw until they are added to image_sequence.

Your challenge

Change the colours, positions and order of the pictures.

2. Make a triangle



A triangle has three sides. Use size for the side length and angle to rotate the whole shape.

Try this in student_images.py

```
my_triangle = triangle(x=-2, y=-2,
                       size=3, angle=45,
                       color="blue", fill="skyblue",
                       speed=5)
```

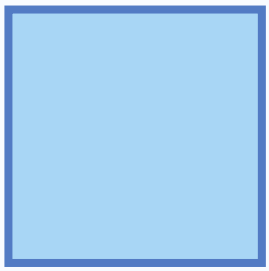
Remember

angle rotates the triangle; it does not change its internal 60-degree angles.

Your challenge

Try angle=0, angle=45 and angle=90.

3. Make a square



A square has four equal sides.

Try this in student_images.py

```
my_square = square(x=-1, y=-1, size=2,
                   angle=20, color="green",
                   fill="lightgreen")
```

Remember

Use size to change both the width and height.

Your challenge

Make three squares with different sizes.

4. Make a rectangle

A rectangle has a width and a height. They can be different.

Try this in student_images.py

```
my_rectangle = rectangle(x=-3, y=-2,
                         width=5, height=2,
                         angle=0, color="blue",
                         fill="lightblue")
```

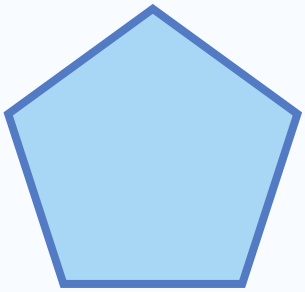
Remember

width controls the horizontal side and height the other side. For this function, use line_width to change the outline thickness.

Your challenge

Make a tall rectangle by switching width and height.

5. Make a pentagon



A pentagon has five sides.

Try this in student_images.py

```
my_pentagon = pentagon(x=-1, y=-2,
                       size=2, angle=0,
                       color="orange", fill="yellow")
```

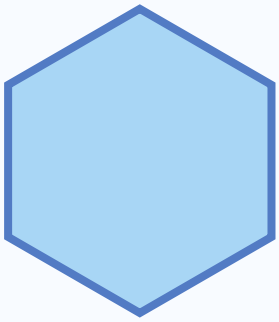
Remember

A regular pentagon has five equal sides.

Your challenge

Try a rotated pentagon with angle=30.

6. Make a hexagon



A hexagon has six sides.

Try this in student_images.py

```
my_hexagon = hexagon(x=-1, y=-2,
                     size=2, color="purple",
                     fill="lavender")
```

Remember

A regular hexagon has six equal sides.

Your challenge

Make a honeycomb design with repeated hexagons.

7. Make a star

The star function draws a five-point star using turns of 144 degrees.

Try this in student_images.py

```
my_star = star(x=-2, y=0, size=3,
               angle=0, color="purple",
               fill="yellow", speed=5)
```

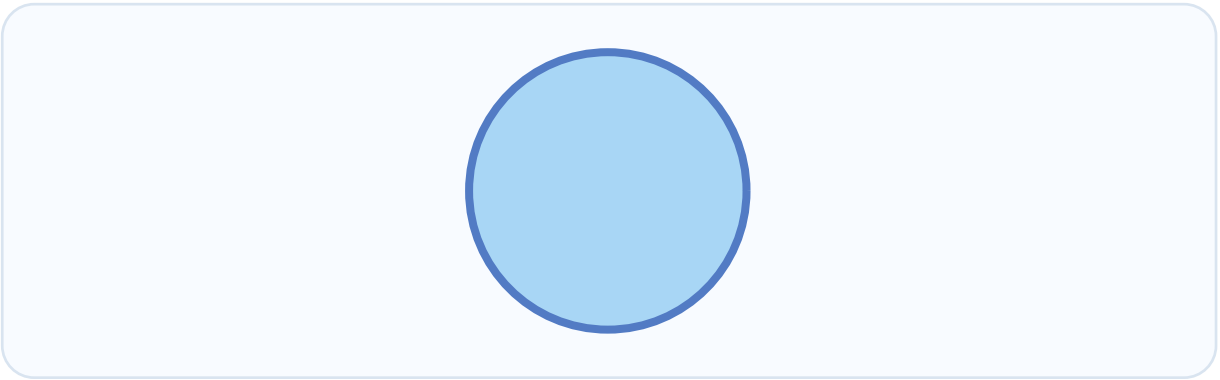
Remember

Change angle to rotate the star. A large star may extend outside the grid.

Your challenge

Make two stars with different colours and angles.

8. Make a circle



A circle is drawn with the curve command behind the scenes.

Try this in student_images.py

```
my_circle = circle(x=0, y=0, radius=1,
                   color="purple", fill="pink",
                   speed=5)
```

Remember

radius controls the distance from the centre to the edge. x and y are the starting point for the drawing, not necessarily the circle centre.

Your challenge

Make a small circle and a large circle.

9. Make a spiral

A spiral uses forward movements that get longer after each turn.

Try this in student_images.py

```
my_spiral = spiral(x=0, y=0,
                   turn=90, steps=12,
                   growth=0.2, color="purple",
                   speed=5)
```

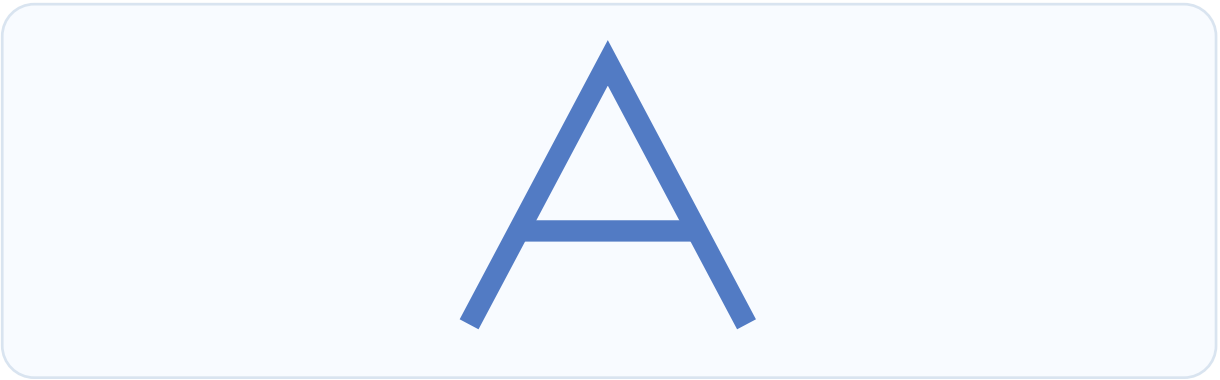
Remember

turn changes the angle each step; growth changes how much longer each line becomes.

Your challenge

Try turn=60 or growth=0.1.

10. Make a letter A



Make a letter A using two sloping lines and a crossbar.

Try this in student_images.py

```
my_a = letter_a(x=-3, y=-2, size=3,
                color="red", speed=5)
```

Remember

The function creates the drawing instructions for you.

Your challenge

Draw your initial in a new colour.

11. Make a letter H

Make a letter H using two vertical lines and one connecting line.

Try this in student_images.py

```
my_h = letter_h(x=-2, y=-2, size=3,
                color="purple", speed=5)
```

Remember

Use size to make the letter taller.

Your challenge

Draw A and H next to each other.

12. Make a curved letter S

The S uses two connected arcs that turn in opposite directions.

Try this in student_images.py

```
my_s = letter_s(x=0, y=2, radius=1,
                color="purple", speed=5)
```

Remember

This uses the corrected curve support in the updated Turtle app. If it draws a circle instead, check that you are running the corrected teacher_sequence.py.

Your challenge

Try a different radius and line width.

13. Make Venetian blinds

A Venetian blind is a stack of thin rectangles (slats).

Try this in student_images.py

```
my_blinds = venetian_blind(
    x=-4, y=-3, width=8,
    slats=7, slat_height=0.55,
    gap=0.3, color="steelblue",
    fill="lightblue", speed=5
)
```

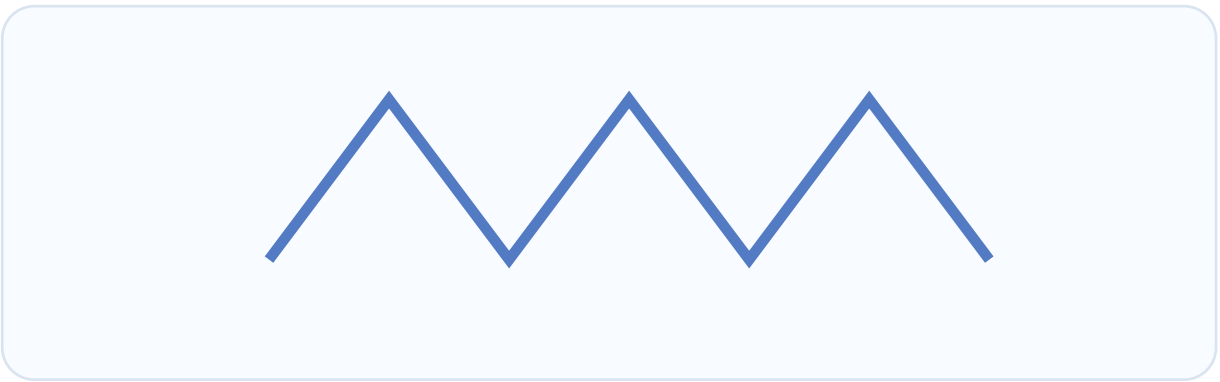
Remember

slats is the number of rectangles; gap is the empty space between them.

Your challenge

Try 5 slats and a larger gap.

14. Make a zigzag



A zigzag changes direction after every segment.

Try this in student_images.py

```
my_zigzag = zigzag(x=-4, y=0,
                   steps=6, segment=1,
                   angle=45, color="orange")
```

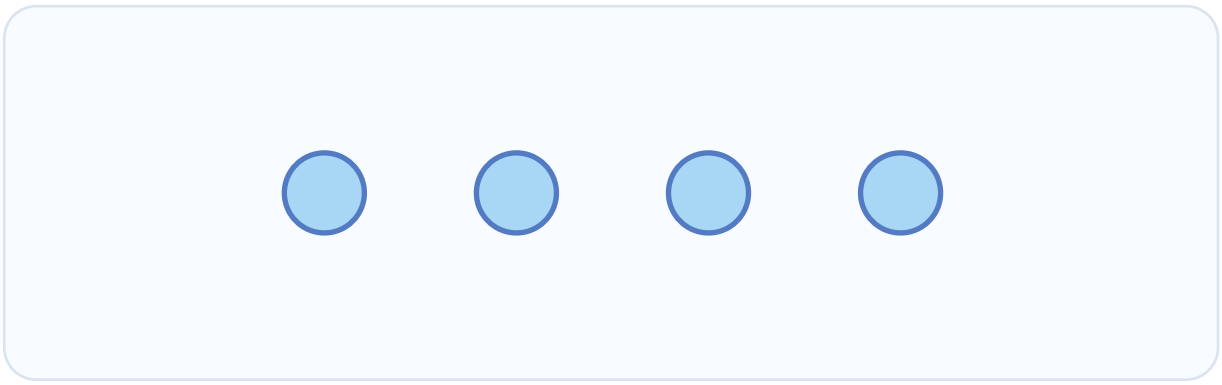
Remember

steps controls how many line segments to draw.

Your challenge

Try steps=8 or angle=30.

15. Repeat a shape



Use repeat_shape to place several copies of one shape.

Try this in student_images.py

```
my_row = repeat_shape(
    circle, count=4,
    x=-4, y=0, dx=2, dy=0,
    radius=0.4, color="blue"
)
```

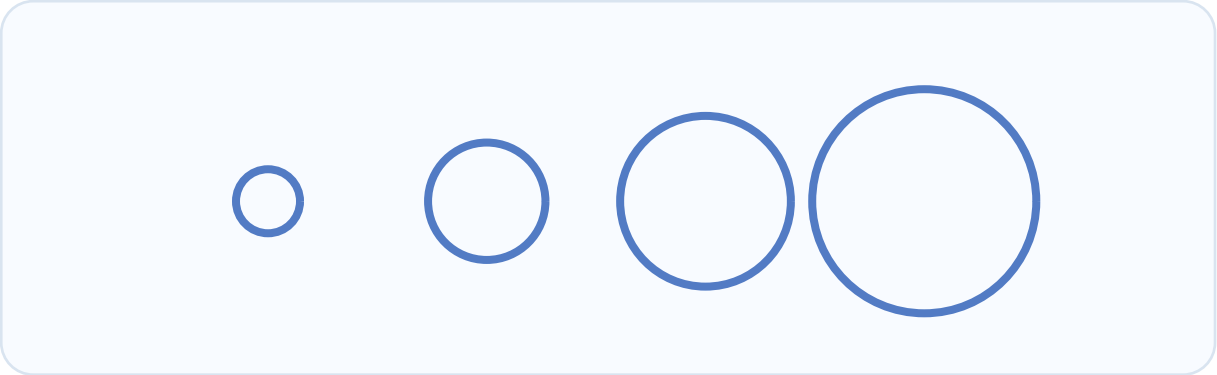
Remember

dx moves each new shape sideways; dy moves it vertically.

Your challenge

Change dy=1 to create a diagonal row.

16. Make growing shapes



Use growing_shapes to draw copies that increase in size.

Try this in student_images.py

```
my_growth = growing_shapes(
    triangle, count=4,
    x=-3, y=-3,
    start_size=1, step=0.5,
    color="green"
)
```

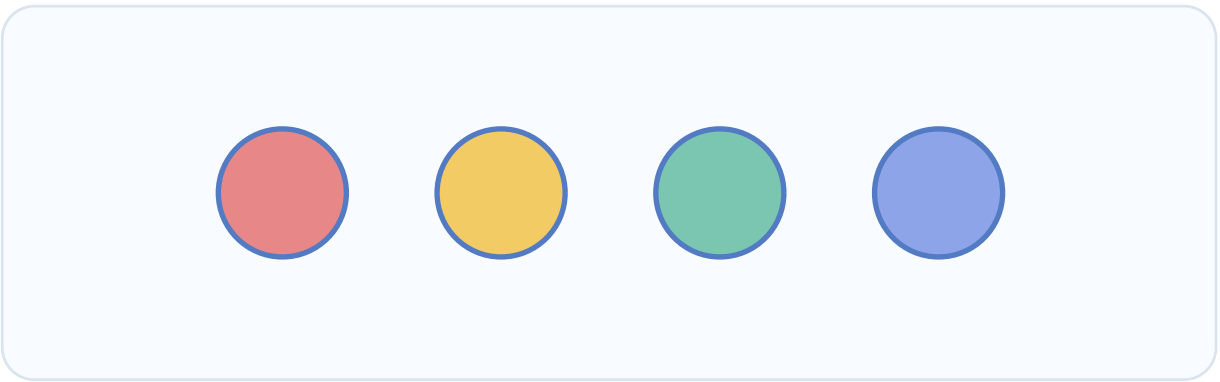
Remember

start_size is the first size; step is the amount added each time. Shapes share a starting position.

Your challenge

Try using square instead of triangle.

17. Use random colours



The library can pick a different colour for each copy.

Try this in student_images.py

```
my_colours = random_colors(
    star, count=4,
    x=-4, y=0, dx=2,
    size=1,
    palette=["red", "blue", "green", "purple"]
)
```

Remember

The colour choices may change each time the program runs.

Your challenge

Try your own palette of four colours.

18. Choose a background

Choose one of the included scenes, or use your own image in the backgrounds folder.

Try this in student_images.py

```
# Put this near the top of student_images.py
background = "beach"

# Other choices:
# "space", "forest", "city", "classroom", "none"

# Or add a PNG/GIF to backgrounds/:
# background = "my_picture.png"
```

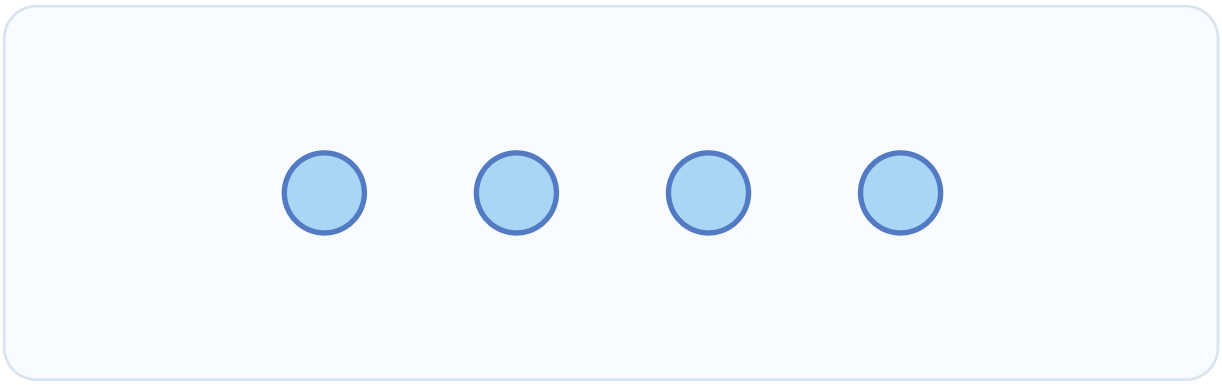
Remember

A PNG or GIF can be used without extra packages. JPEG requires Pillow. The background sits behind the drawing.

Your challenge

Make a beach scene with a yellow circle for the sun.

19. Put your creations together



Create a whole scene by combining the library functions into one picture.

Try this in student_images.py

```
background = "space"
from turtle_shapes import *

sun = circle(x=2, y=2, radius=0.7,
             color="orange", fill="yellow")
stars = random_colors(
    star, count=3, x=-4, y=1, dx=1.5,
    size=0.7, palette=["yellow", "white"]
)
planet = circle(x=-1, y=-2, radius=1,
                color="purple", fill="pink")

image_sequence = [
    ("Sun", sun),
    ("Stars", stars),
    ("Planet", planet),
]
```

Remember

Add only the picture variables you have actually created. Existing Pattern() lessons can still be used in the same app.

Your challenge

Make your own scene with at least three different functions.